Byeonghwa Jeong

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Education

Ph.D. **University College London**, Geography (Geospatial Data Science) — London, UK
Sept 2019 – June 2023
- Thesis: An Integrated Deep Learning Model Using Retail, Residential and Footfall Change Data to Predict Retail Gentrification.
- Supervisors: Professor Paul Longley and Professor James Cheshire.

M.Sc. **University of Edinburgh**, Geographical Information Science — Edinburgh, UK
Sept 2017 – Aug 2018
- Graduated with Distinction.
- Dissertation: Anticipation of Commercial Gentrification through Time and Space: Seoul, South Korea.

B.A. **Kyungpook National University**, Geography — Daegu, South Korea
Mar 2008 – Aug 2015

Experience

Oak Ridge National Laboratory, R&D Associate — Oak Ridge, Tennessee, USA
Feb 2025 – present
1 year 9 months

Develop geospatial AI methods for population and socioeconomic measurement, disaster impacts, and infrastructure risk.
- Develop geospatial AI methods for large-scale population modelling using graph neural networks, computer vision, satellite embeddings, and spatial machine learning.
- Develop a deep-learning framework using AlphaEarth satellite embeddings to reconstruct gridded population surfaces while explicitly modelling populated and zero-population areas and evaluating geographic transferability across Taiwan, Ecuador, the Democratic Republic of the Congo, and Japan.
- Train and evaluate PyTorch deep-learning and graph models on NVIDIA A100 and H200 GPU systems, focusing on scalable architectures, geographic transferability, and efficient modelling of large spatial datasets.
- Contribute to LandScan Global and LandScan Mosaic, including global gridded-population production, rapid population-impact estimation for disasters and conflicts, and development of population-model inputs.
- Lead or co-lead research on multi-scale population forecasting, satellite-based population reconstruction, building-level wealth estimation, and hurricane-induced power-outage forecasting.
- Submitted a $228,000 competitive ORNL seed proposal as Principal Investigator on graph-based outage and population-impact prediction.

University of Toronto, School of Cities, Postdoctoral Research Fellow — Toronto, Ontario, Canada
July 2023 – Jan 2025
1 year 7 months

Developed computational methods for building-level and downtown-scale recovery analysis using privacy-preserving footfall, real-estate, socioeconomic, and open geospatial data.
- Estimated sampling uncertainty, imputed missing footfall observations, developed data-driven downtown boundaries, and applied clustering and explainable machine learning to investigate heterogeneous recovery patterns.
- Held day-to-day responsibility for training a graduate research intern on an LLM-based urban-planning chatbot project from research design and coding through manuscript development.
- Provided project-based computational training and iterative feedback to undergraduate research interns.

Daejeon Sejong Research Institute, External Collaborative Researcher — Daejeon, South Korea
Big data-driven strategies for enhancing the Science City Daejeon brand. — Oct 2024 – Dec 2024 — 3 months
- Collected and analysed web and social-media text using TF-IDF, topic modelling, and network analysis.
- Contributed to data collection, preprocessing, interpretation, and policy-report preparation.

Daejeon Sejong Research Institute, External Collaborative Researcher — Daejeon, South Korea
Research on regional identity, civic participation, and place-based assets using big-data analysis. — Aug 2023 – Oct 2023 — 3 months
- Collected and preprocessed web and social-media text and applied TF-IDF, topic modelling, and network analysis.

Daejeon Sejong Research Institute, External Technical Consultant — Daejeon, South Korea
Technical consulting for social-media data collection and text mining. — July 2022 – Sept 2022 — 3 months
- Advised on Python web scraping, text preprocessing, topic modelling, and network analysis.

Korea Environment Institute, Part-time Modeller — Sejong, South Korea
Developed a deep-learning spatial interpolation model for atmospheric CO2 estimation. — Aug 2021 – Oct 2021 — 3 months
- Combined OCO-2 satellite observations and GEOS-Chem outputs to estimate unobserved atmospheric CO2 values.

Korea Research Institute for Human Settlements, Assistant Research Fellow — Sejong, South Korea
Developed national territorial-monitoring indicators and spatial accessibility measures. — Sept 2018 – Sept 2019 — 1 year 1 month
- Developed national grid-based indicators of access to schools, parks, and other daily infrastructure.
- Constructed longitudinal indicators of building age for territorial monitoring and policy analysis.

Kyungpook National University, Student Research Assistant — Daegu, South Korea
Research assistance in remote sensing, urban carbon, and geospatial analysis. — Nov 2015 – June 2016 — 8 months
- Analysed urban carbon patterns from OCO-2 satellite data and developed foundational skills in Python, geostatistics, and remote-sensing data processing.

Publications

Development of Grid-Based Population Projection Method: A Modified Cohort Component Approach Applied to South Korea — Jan 2026
Bokyeong Lee, *Byeonghwa Jeong*
10.1002/psp.70161 (Population, Space and Place)

Popnet: Computer Vision Based Deep Learning Model for Forecasting Gridded Population — Jan 2026
Byeonghwa Jeong, Bokyeong Lee
10.1080/13658816.2025.2514792 (International Journal of Geographical Information Science)

A Large Language Model-Based Chatbot System Framework for Urban Planners — Jan 2025
Xiaoxin Zhou, *Byeonghwa Jeong*, Karen Chapple
10.1016/j.eswa.2025.130307 (Expert Systems with Applications)

A First Open Retail Boundary Dataset for South Korea Using Open Data and Computer Vision — Jan 2025
Byeonghwa Jeong, Juhwan Song, Younghoon Kim
10.1038/s41597-025-04958-1 (Scientific Data)

Does Class Matter? Understanding Differential Pandemic Recovery via a Building Typology Jan 2025
Byeonghwa Jeong, Karen Chapple
10.1177/23998083251326128 (Environment and Planning B: Urban Analytics and City Science)

Assessing Downtown Recovery Rates and Determinants in North American Cities After the COVID-19 Pandemic Jan 2024
Amir Forouhar, Karen Chapple, Jeff Allen, *Byeonghwa Jeong*, Julia Greenberg
10.1177/00420980241270987 (Urban Studies)

A New Commercial Boundary Dataset for Metropolitan Areas in the USA and Canada, Built from Open Data Jan 2024
Byeonghwa Jeong, Jeff Allen, Karen Chapple
10.1038/s41597-024-03275-3 (Scientific Data)

Topic Modeling of the Transition of the Sense of Place in an Urban Regeneration Area: A Case Study of Daegu Bangcheon Market Jan 2020
Byeonghwa Jeong, Junwoo Kim
(Journal of Daegu Gyeongbuk Studies)

Application and Evaluation of Self-Organising Map-Based Spatial Clustering for Regional Policy Recommendations Jan 2019
Byeonghwa Jeong, Junwoo Kim
(Journal of the Korean Geographical Society)

Comparative Evaluation of South versus North Goseong County in Terms of Forest Carbon Offset Potential Jan 2016
Byeonghwa Jeong, Jung-Sup Um
(Spatial Information Research)

Manuscripts Under Review

- Jeong, B., Lebakula, V., and Urban, M. (2026). From grids to communities: A graph neural network framework for multi-scale population forecasting. AAAI AI for Social Impact (under review).
- Jeong, B., Lee, S., and Myers, A. (2026). Dual-structure spatio-temporal graph learning for hurricane-induced power outage forecasting. AAAI AI for Social Impact (under review).
- Jeong, B. and Kim, J. (2026). Predicting retail gentrification through composite retail vitality index and spatial competition metrics. International Journal of Urban Sciences (under review).
- Jeong, B., Tuccillo, J., and Gaboardi, J. (2026). Building-level wealth index prediction using DHS survey data and machine learning: A flexible multi-scale framework. Computers, Environment and Urban Systems (under review).

Current Research Projects

- Interpretable population estimation from aerial imagery - Developing a two-stage semantic-segmentation and regression framework with ensemble self-training, pseudo-label filtering, temporal transfer, and landscape-class contribution maps; manuscript submission planned in 2026.
- Population surface reconstruction using AlphaEarth embeddings - Developing a deep-learning framework that uses AlphaEarth satellite embeddings to reconstruct gridded population distributions while explicitly learning both populated and zero-population areas, with geographic-transfer evaluation across Taiwan, Ecuador, the Democratic Republic of the Congo, and Japan.
- Scalable latent global routing for large-scale spatio-temporal graph learning - Developing ST-LAG, a latent-mediated graph architecture that combines sparse local graph propagation with node-to-latent global routing to model non-local dependencies without dense N x N node interactions.
- Large-scale graph learning for population dynamics - Applying ST-LAG to continental-scale gridded population forecasting across the United States over more than 150,000 spatial units.
- Scalable regional power-outage forecasting - Applying ST-LAG to hurricane-induced power-outage forecasting across the southeastern United States and developing an API for integration into ORNL's EDGAR platform.

Technical Reports

- Jeong, B., Lebakula, V., and Urban, M. (2026). PopGNN: Graph neural network-based flexible future population forecasting model. Oak Ridge National Laboratory.
- Jeong, B., Adams, D., Zimmer, A., and Urban, M. (2026). Satellite embedding-based population imputation for areas with missing building-footprint data: A computer vision-based approach. Oak Ridge National Laboratory.

Policy Reports

- Chapple, K., Forouhar, A., Jeong, B., and Ortynsky, S. (2025). Assessing the doom loop. Lincoln Institute of Land Policy.
- Joo, H., Han, S., Jeong, E., Jeong, B., Song, J., and Noh, M. (2025). Big data-driven strategies for enhancing the 'Science City Daejeon' brand. Daejeon Sejong Research Institute.
- Joo, H., Noh, M., and Jeong, B. (2024). Basic research for citizen-participatory urban branding. Daejeon Sejong Research Institute.
- Joo, H., Han, S., Yoon, S., Moon, J., and Jeong, B. (2023). Discover 'regional things' using big-data analysis and civic collaboration. Daejeon Sejong Research Institute.
- Lim, E., et al., including Jeong, B. (2021). Development and utilisation of a multiple deprivation index for place-based living-facilities policy. Korea Research Institute for Human Settlements.
- Lim, E., et al., including Jeong, B. (2019). National Territorial Monitoring Report 2018. National Geographic Information Institute.

Datasets

- Lebakula, V., Gonzales, J., Stipek, C., Tsybina, E., Zimmer, A., Nukavarapu, N., Jeong, B., Reynolds, B., Kaufman, J., Fan, J., Martin, A., Buck, W., Basford, S., Faxon, A., Meade, S., and Urban, M. (2025). LandScan Global 2024. Oak Ridge National Laboratory. Dataset. https://doi.org/10.48690/1532445

Research Funding and Proposals

- Principal Investigator. A Graph Neural Network Approach for Forecasting Disaster-Driven Power Outages and Population Impacts. ORNL competitive internal Seed programme. $228,000 requested; submitted 2026; decision pending.
- Project contributor. Marshalling a Robust Downtown Recovery for Ottawa. Canadian Urban Institute. $109,514 equivalent. Contributions: data curation and preprocessing, methodological selection, analysis, visualisation, and proposal writing.
- Project contributor. Assessing the Doom Loop: Scenarios in Four North American Downtowns. Lincoln Institute of Land Policy. $52,931 equivalent. Contributions: data collection and preprocessing, explainable machine learning, analysis, and proposal writing.
- Project contributor. Evaluation and Characteristics of Coastal Areas Using Image Segmentation. Korea Maritime Institute. $2,885 equivalent. Contributions: idea development, provision of core data, writing, and review.

Teaching, Research Training, and Student Development

- Guest lecture (Mar 2025): Computer-vision methods for delineating regional boundaries. Center for Territorial Monitoring, Korea Research Institute for Human Settlements.
- Guest lecture (Aug 2024): Applications of AI for predicting social phenomena. Department of Geography Education, Korea National University of Education.
- Guest lecture (Jun 2023): Social phenomena and spatial AI. Department of Geography, Kyungpook National University.
- Primary day-to-day research mentor for Xiaoxin Zhou, graduate research intern, University of Toronto, providing sustained training from research-question development and coding through academic writing and manuscript revision; the project resulted in a peer-reviewed article in Expert Systems with Applications for which I served as corresponding author.
- Provided project-based computational training to undergraduate research interns Christopher Kim and Brandon Choi, including analysis using project code and data, review of analytical outputs, and iterative feedback on coding and research communication.
- Initiated and led a collaborative research project with doctoral researchers Gaweon Shin and Changwha Oh, coordinating research design, computational methods, analytical workflows, and research communication.
- Mentor to a doctoral student at the University of North Carolina at Charlotte through the 2026-2027 KAGES Mentoring Program, focusing on workload management and job-application preparation.

Conference Presentations

- 2026 - Building-level wealth index prediction using DHS data and machine learning. American Association of Geographers Annual Meeting.
- 2024 - Delineating commercial boundaries using an AI-based computer-vision technique with open data in South Korea. AsiaCarto 2024.
- 2024 - Typology of downtown recovery at the granular scale. Geoinformatics 2024.

Professional Service

- Program Committee Member, 6th International Workshop on Big Data and AI-Assisted Tools, Methods, and Use Cases for Innovative Scientific Discovery, in conjunction with the 2026 IEEE International Conference on Big Data.
- Member of the Organising Committee, International Conference on Geospatial Information Science: Spatially Enabled Society with AI and Digital Twin (2019).
- Journal reviewer: International Journal of Geographical Information Science (2025).
- Journal reviewer: Computers and Electrical Engineering (2026).
- Journal reviewer: Papers in Regional Science (2026).
- Journal reviewer: Spatial Information Research (2026).
- Journal reviewer: Transactions in GIS (2026).
- Journal reviewer: Applied Spatial Analysis and Policy (2026).
- Founded and currently run Geo-opportunity (@geo_opportunity), an Instagram account that curates and shares PhD and postdoctoral opportunities in geography and GIS for students and early-career researchers.

Honours and Prizes

- 2020 - Best Paper Award, Daegu Gyeongbuk Development Institute. Topic Modeling of the Transition of the Sense of Place in an Urban Regeneration Area.
- 2015 - Outstanding Paper Award, Korea Spatial Information Society. Comparative Evaluation of Pixel-Value Signatures between UN A/R CDM Project Areas in South and North Goseong.
- 2013 - June-Myeong Scholarship for ArcGIS proficiency, GEO C&I Corporation. $300.

Technical Skills

Programming and data engineering: Python (pandas, GeoPandas, NumPy, PySpark), SQL/MariaDB, R, web scraping, reproducible geospatial workflows

Machine learning and AI: PyTorch, scikit-learn, XGBoost, computer vision, graph neural networks, spatio-temporal modelling, NLP, large language models, retrieval-augmented generation, prompt engineering, model explainability

GIS and remote sensing: ArcGIS, QGIS, Google Earth Engine, raster and vector analysis, OpenStreetMap, geospatial databases, spatial statistics, network analysis

Computing: HPC/GPU workflows

Languages

Korean: Native

English: Professional proficiency